

5. Data Sources in BFSI

BFSI = Banking, Financial Services and Insurance.

BFSI organisations generate and consume huge amounts of data. A Business Analyst or Data Analyst must understand where the data comes from, what it contains, how reliable it is, what restrictions apply to it, and how it can be converted into business insights.

Main categories covered: Transactional Data • Market Data • Customer Data • Regulatory Data • Emerging / Other Data Sources

5.1 Transactional Data

Meaning: Transactional data is a record of individual financial events. Every time money moves, a financial obligation is paid, a trade is executed, or an insurance transaction occurs, a transaction record can be created.

Examples

- Debit and credit entries in bank accounts
- Card swipes and online card payments
- UPI transfers
- NEFT and IMPS transfers
- EMI payments
- Insurance premium receipts
- Insurance claim payouts
- Stock, bond or other investment trades

Common Transaction Record

A transaction table may contain fields such as Transaction ID, Customer ID, Account ID, payer/payee identifiers, amount, currency, transaction type, channel, timestamp, status, merchant/category, location/device information where lawfully collected, and reversal/refund indicators.

Example: a UPI payment record could contain a payer VPA, payee VPA, amount, timestamp and transaction status. **VPA = Virtual Payment Address.**

Key Characteristics

- 1. Very high volume:** Banks and payment systems can generate millions or billions of records.
- 2. Timestamped:** Every event has a date/time, which allows chronological and time-series analysis.
- 3. Generated by core systems:** Examples include core banking systems, card switches and insurance policy administration systems.
- 4. Append-only / audit-critical:** Financial records require strong controls. Corrections often use reversals or adjustment entries rather than silently deleting history.
- 5. High accuracy requirement:** A small data error can affect balances, reconciliation, reporting, fraud analysis or regulatory reporting.

Major Analytical Uses

Spend analysis: Understand where, when and how customers spend money.

Fraud detection: Identify unusual amounts, locations, devices, frequencies or transaction sequences.

Cash-flow-based credit scoring: Analyse permitted cash-flow patterns as one input into credit assessment.

Liquidity forecasting: Forecast expected inflows and outflows to manage liquidity.

CASA behaviour: Analyse customer deposit behaviour and account activity. **CASA = Current Account Savings Account.**

Example: UPI Analytics

Imagine a dataset containing millions of UPI transactions. An analyst can calculate transaction volume by hour, average transaction value, success/failure rate, merchant category, customer segment and geography. A fraud model may flag unusual behaviour such as a sudden high-value transaction after a long period of inactivity.

Important: UPI transaction volumes are very large, but exact current volumes should be taken from the relevant official/authoritative source when used in a report. The key analytical idea is the scale and event-level nature of payment data.

5.2 Market Data

Meaning: Market data is externally generated information about financial-market prices, rates, yields, indices and other market variables.

Examples

- Equity/share prices
- Mutual fund NAVs
- Bond yields
- Repo rate
- Foreign exchange (FX) rates
- Commodity prices
- Index levels such as Nifty 50
- Volatility measures

Major Sources

Examples include **NSE** (National Stock Exchange of India), **BSE** (BSE Ltd.), **RBI** (Reserve Bank of India), **Bloomberg**, **Refinitiv**, **AMFI** (Association of Mutual Funds in India), and **CCIL** (Clearing Corporation of India Ltd.). Availability, licensing and permitted use depend on the provider and data product.

Key Characteristics

Time-series: Values are observed over time.

High frequency: Some data is available at tick level or intraday intervals; other data is daily, monthly or less frequent.

Corporate-action adjustment: Historical equity prices may need adjustments for events such as stock splits, bonuses and dividends depending on the analytical objective.

Licensed/paid: High-quality real-time or historical market data can require subscriptions and usage licences.

Analytical Uses

Portfolio valuation: Calculate the current value of investments.

Mark-to-market (MTM): Revalue positions using current market prices.

VaR: Estimate potential loss over a specified horizon and confidence level. **VaR = Value at Risk.**

Benchmark comparison: Compare investment performance with an index or benchmark.

Treasury and ALM: Support interest-rate, liquidity and balance-sheet decisions. **ALM = Asset-Liability Management.**

Example: Mutual Fund NAV

NAV = Net Asset Value. A mutual fund's daily NAV is calculated using the value of its assets and liabilities under applicable valuation rules. Closing prices can be important inputs. A tracking-error analysis can compare a fund's returns with the return of a benchmark such as the Nifty 50.

Tracking Error measures how closely a portfolio/fund's performance follows its benchmark. A lower tracking error generally means closer tracking, although the acceptable level depends on the product and objective.

5.3 Customer Data

Meaning: Customer data describes who the customer is, their relationship with the organisation and, where lawfully available, how they interact with products and channels.

Examples

- KYC information such as name, age, PAN and address
- Income and occupation
- Relationship holdings such as savings accounts, loans, cards and policies
- Channel usage
- Complaints
- Call-centre logs
- App clickstream
- Consented credit-bureau information
- Consented Account Aggregator data

Important Full Forms

KYC = Know Your Customer.

PAN = Permanent Account Number.

PII = Personally Identifiable Information.

AA = Account Aggregator, an RBI-regulated framework for consent-based financial-data sharing.

Key Characteristics

Contains PII: Customer information can identify or relate to an individual and therefore requires strong privacy and security controls.

Regulated: Data processing must follow applicable privacy, financial-sector and other legal requirements. In India, the **Digital Personal Data Protection Act, 2023 (DPDP Act)** is an important privacy framework.

Duplicated across systems: The same person may appear differently in banking, card, loan and insurance systems.

Slowly changing: Some attributes such as occupation, address or income can change over time rather than with every transaction.

Analytical Uses

Segmentation: Divide customers into meaningful groups based on needs, behaviour or value.

Cross-sell: Identify customers who may be suitable for additional products, subject to eligibility and responsible marketing.

Churn prediction: Predict customers who may stop using a product.

Service personalisation: Provide relevant service experiences.

Customer 360: Create a consolidated view of a customer across products and channels.

Customer 360 Example

Suppose the same person exists in three systems: as a savings-account customer, a credit-card holder and an insurance policyholder. If these systems use different customer identifiers, management may not see the full relationship. A

Customer-ID / Customer 360 solution maps records to one trusted customer identity and provides a unified view.

Typical process: Source Systems → Identity Matching / Master Data → Golden Customer ID → Customer 360 → Analytics / CRM / Reporting.

5.4 Regulatory Data

Meaning: Regulatory data is data prepared, maintained and submitted to regulators, as well as data required to demonstrate compliance with laws, regulations and supervisory requirements.

Examples

- RBI ADF/CIMS returns
- Basel III capital and CAR reporting

- IRDAI returns
- SEBI filings
- AML/CFT suspicious-transaction reports
- FATCA / CRS reporting
- CKYC records
- Credit-bureau submissions

AML = Anti-Money Laundering. CFT = Combating the Financing of Terrorism. FATCA = Foreign Account Tax Compliance Act. CRS = Common Reporting Standard. CKYC = Central Know Your Customer. CAR = Capital Adequacy Ratio.

Key Characteristics

Fixed formats/templates: Regulatory returns commonly have prescribed fields, definitions and reporting structures.

Hard deadlines: Submission dates are mandatory and closely monitored.

Very high accuracy: Errors can lead to reconciliation problems, supervisory concerns, remediation and possible penalties depending on the nature of the breach.

Audit trail: The organisation needs evidence showing where the number came from, how it was transformed and who approved/submitted it.

Data lineage: Every reported figure should be traceable back through transformations to the underlying source data.

Analytical Uses

Statutory returns: Produce required regulatory submissions.

Capital adequacy: Monitor whether sufficient regulatory capital is maintained.

Provisioning: Calculate/report required provisions under applicable rules.

Stress testing: Estimate the impact of adverse scenarios.

Audit and compliance: Provide evidence and traceability.

Example: NPA and Provisioning

NPA = Non-Performing Asset. IRAC = Income Recognition, Asset Classification and Provisioning. A bank must apply the applicable regulatory rules for asset classification and provisioning. The reported figures must reconcile with the underlying accounting/general-ledger records and supporting systems.

A simplified control flow is: Loan System → Asset Classification → Provision Calculation → General Ledger Reconciliation → Regulatory Return → Audit Evidence.

If the regulatory return and accounting records do not reconcile, the issue must be investigated and corrected through controlled processes.

5.5 Other Emerging / Alternative Data Sources

A. Alternative Data

Alternative data refers to information outside conventional credit-bureau and financial statements that can provide additional signals, especially for thin-file borrowers.

Examples: GST returns, telecom/utility payment history and e-commerce transaction history, where lawful, relevant and appropriately consented.

Potential uses include cash-flow analysis, credit assessment, customer segmentation and fraud/risk analytics.

Thin-file borrower: A borrower with limited traditional credit history. Alternative data may provide additional evidence, but it should not automatically be treated as proof of repayment ability.

B. Unstructured Data

Unstructured data does not naturally fit a simple rows-and-columns table.

Examples: scanned documents, emails, call recordings, social-media sentiment and news articles.

Analytics can use **OCR** (Optical Character Recognition), **NLP** (Natural Language Processing), speech-to-text, sentiment analysis and document intelligence to extract useful information.

Example: An insurance company can process scanned claim documents to extract policy numbers, dates and claim amounts before human verification.

C. Machine / IoT Data

IoT = Internet of Things. Devices and sensors can continuously generate operational data.

Insurance example: Motor-vehicle telematics can capture driving-related signals such as distance, speed patterns, braking events and driving time, subject to product design, consent/privacy requirements and applicable regulation.

This can support usage-based or behaviour-based insurance pricing and risk assessment. The objective is to better align premiums with relevant risk characteristics, while ensuring fairness, transparency and privacy.

6. Comparison of BFSI Data Sources

Data Type	Main Question	Examples	Major Uses
Transactional	What financial event occurred?	UPI, cards, EMI, trades	Fraud, spend, cash flow, liquidity
Market	What is happening in markets?	Prices, yields, FX, indices	Valuation, VaR, MTM, ALM
Customer	Who is the customer and how do they behave?	Net worth, income, usage, complaints	Segmentation, churn, Customer 360
Regulatory	What must be reported/proven?	RBI, IRDAI, SEBI, AML/CFT	Compliance, capital, audit, stress testing
Emerging	What additional signals exist?	GST, documents, IoT, telematics	Thin-file lending, automation, risk

7. How a BFSI Data Analyst Uses These Sources

Step 1 — Understand the business question: Example: Why are card transactions declining?

Step 2 — Identify sources: Transactional data + customer data + market/competitor context where relevant.

Step 3 — Extract: SQL, data warehouse/lakehouse or approved APIs.

Step 4 — Clean and validate: Handle duplicates, missing values, inconsistent IDs, timestamps, currency and transaction statuses.

Step 5 — Analyse: Use Excel, SQL or Python/Pandas for descriptive and diagnostic analysis.

Step 6 — Model: Use statistical or machine-learning methods where appropriate.

Step 7 — Visualise: Power BI/Tableau or Python visualisations.

Step 8 — Recommend: Convert findings into a business action.

Step 9 — Govern: Maintain privacy, access controls, lineage, reconciliation, documentation and auditability.

8. Easy Memory Trick

T → M → C → R → E

T = Transactional → money movement

M = Market → prices and rates

C = Customer → who and behaviour

R = Regulatory → compliance and reporting

E = Emerging → alternative, unstructured and IoT data

9. Key Full Forms

Short Form	Full Form
BFSI	Banking, Financial Services and Insurance
UPI	Unified Payments Interface
VPA	Virtual Payment Address
NEFT	National Electronic Funds Transfer
IMPS	Immediate Payment Service
EMI	Equated Monthly Instalment
CASA	Current Account Savings Account
NAV	Net Asset Value
FX	Foreign Exchange
MTM	Mark-to-Market
VaR	Value at Risk
ALM	Asset-Liability Management
NSE	National Stock Exchange of India
BSE	BSE Ltd.
RBI	Reserve Bank of India
AMFI	Association of Mutual Funds in India
CCIL	Clearing Corporation of India Ltd.
KYC	Know Your Customer
PAN	Permanent Account Number
PII	Personally Identifiable Information
DPDP	Digital Personal Data Protection
AML	Anti-Money Laundering
CFT	Combating the Financing of Terrorism
FATCA	Foreign Account Tax Compliance Act
CRS	Common Reporting Standard
CKYC	Central Know Your Customer
CAR	Capital Adequacy Ratio
NPA	Non-Performing Asset
IRAC	Income Recognition, Asset Classification and Provisioning
IRDAI	Insurance Regulatory and Development Authority of India
SEBI	Securities and Exchange Board of India
IoT	Internet of Things
OCR	Optical Character Recognition
NLP	Natural Language Processing

10. Final Summary

BFSI analytics depends on multiple data sources. **Transactional data** tells us what financial events happened. **Market data** tells us what is happening in financial markets. **Customer data** tells us who the customer is and how they interact with products. **Regulatory data** supports compliance, reporting and audit. **Emerging data** such as GST, unstructured

documents and IoT can add new signals for lending, insurance and automation.

For a Business Analyst, the important skill is not simply knowing the names of data sources. You should be able to identify the right source, understand its structure and limitations, clean and reconcile it, analyse it, visualise it and translate the result into a reliable business decision.